6.1 TRANSFORMER-INTRODUCTION

A Transformer is a static device which transfers electrical energy from one circuit to another circuit at the same frequency. It works on the principle of mutual induction. The transformer works only on A.C supply.

Generating stations generate electricity at a voltage of 11KV. The electric power from the generating station is to be brought to the consumers end from 33KV, 66KV etc through various transmission stages.The transformer is used to step down (or) step up the voltage required according to the requirement as shown in fig. 6.1.



Fig 6.1 *Transformer in Transmission Line.*

6.2 CONSTRUCTION OF TRANSFORMER CORE

Based on the construction, it is classified into 3 types. They are:

i. Core type

ii. Shell type and

iii. Berry type

6.2.1 Core Type

The winding surrounds the core is called core type transformer. The Figure 6.2 represents the core type.

The magnetic circuit is made up of laminated iron core. Silicon steels are used to reduce the hysteresis loss in the iron core, also laminated sheets are insulated from one another by a layer of varnish insulation.

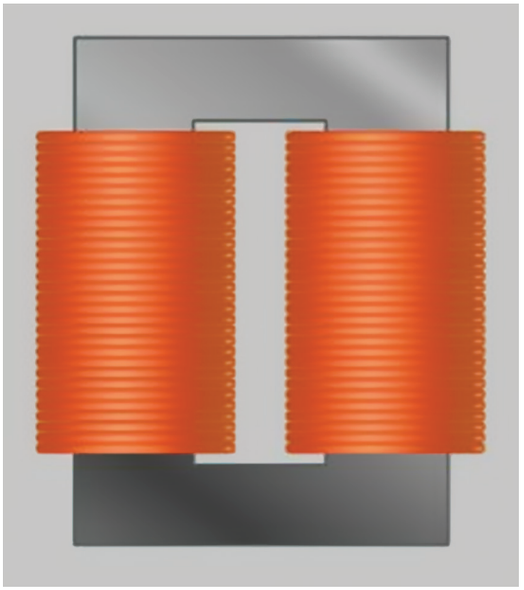


Fig 6.2 *Core Type Construction*

Advantages

i) Easily handled and maintained.

ii) The coils are in the outside, so it will get cooled easily.

Disadvantages

i) High magnetic loss.

ii) High leakage flux.

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